SCAN Foundations

PSYCH 511.001 Spring 2026

2026-01-12

Foundations of Social, Cognitive, and Affective Neuroscience (SCAN)

Instructor

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Where & When

467 Moore Thursdays, 1-4 PM

About the course

The first scientific psychologists were physiologists fascinated by the possibility of understanding the mind by studying the brain. In this course, we will explore the historical roots and contemporary challenges associated with the study of biological approaches to complex adaptive behavior. In doing so, we will read and examine critically primary source readings that discuss basic patterns and processes of brain structure and function. The goal is to provide students with a basic foundation of knowledge about the structures and functions of the nervous system that can provide the basis for future study.

This course is one of two required courses for the Specialization in Cognitive and Affective Neuroscience (SCAN).

Prerequisites

Undergraduate coursework in neuroscience or physiological psychology such as the equivalents of PSYCH 260 or BIO 469/470.

Week 01: Jan 12-16

Thu, Jan 15

Topic(s)

- Structure of the course
- Levels of analysis
- Causality in brain and behavior
- Does neuroscience need behavior? What does psychology need?

Readings

- Required:
  - Krakauer, Ghazanfar, Gomez-Marin, MacIver, & Poeppel (2017)
  - Parada & Rossi (2018)
  - Newcombe (2017)
- Optional:
  - Siddiqi, Kording, Parvizi, & Fox (2022)
  - Churchland & Sejnowski (1988)
  - Favela (2020)
  - Ross & Bassett (2024)

Materials

- Slides | [PDF]

Assignments

- *Assigned*: Exercise LVL • Levels/Terms. *Due*: 2026-01-22. | Canvas dropbox |

Week 02: Jan 19-23

Thu, Jan 22

Topics

- Neuroanatomy

Readings

- Neuroanatomy notes

Watch

- Leicester Medical School Radiology (2021) (Through ~ 8:20)

Materials

- Slides. PDF
- Allen Brain Atlas.

Assignments

- *Due*: Exercise LVL • Levels/Terms. Canvas dropbox
- *Assigned*: Exercise ANAT • Neuroanatomy. *Due*: 2026-01-29. | Canvas dropbox |

Week 03: Jan 26-30

Thu, Jan 29

Topic(s)

- Action

Reading(s)

- Shenoy, Sahani, & Churchland (2013)
- Goulding, Bollu, & Büschges (2025)
- Recommended:
  - *Hardwired for song and dance* (2023)

Materials

- Action notes

Assignments

- *Due*: Exercise ANAT • Neuroanatomy. | Canvas dropbox |

Week 04: Feb 2-6

Thu, Feb 05

Topic(s)

- Perception

Readings

- Khalsa et al. (2018)
- Shenoy et al. (2013)

Materials

- Perception notes

Assignment(s)

- *Assigned*: Final project proposal. *Due*: 2026-03-05. | Canvas dropbox |
- *Assigned*: Exercise PA • Perception & Action. *Due*: 2026-02-12. | Canvas dropbox |

Week 05: Feb 9-13

Thu, Feb 12

Topic(s)

- Learning & memory

Readings

Materials

Assignment(s)

- *Due*: Exercise PA • Perception & Action. | Canvas dropbox |

Week 06: Feb 16-20

Thu, Feb 19

Topic(s)

- Emotion

Readings

- Malezieux, Klein, & Gogolla (2023)
- Watabe-Uchida, Eshel, & Uchida (2017)
- Siegel et al. (2018)
- Recommended
  - LeDoux (2000)

Materials

- Notes

Assignment(s)

- *Assigned*: Exercise EMO • Measuring emotion. *Due*: 2026-02-26. | Canvas dropbox |

Week 07: Feb 23-27

Thu, Feb 26

Topic(s)

- Sleep & circadian rhythms

Readings

- Nir & Tononi (2010)
- Rasch & Born (2013)

Materials

Assignment(s)

- *Due*: Exercise EMO • Measuring emotion. | Canvas dropbox |
- *Assigned*: Exercise ZZ • Sleep & Circadian Rhythms. *Due*: 2026-03-05. | Canvas dropbox |

Readings

Materials

Week 08: Mar 2-6

Thu, Mar 5

Topic(s)

- Evolution of the nervous system

Readings

- Required:
  - Charvet & Finlay (2012)
  - Hofman (2014)
  - Sachkova, Modepalli, & Kittelmann (2025)
- Optional:
  - Castrillon et al. (2023)

Materials

- Notes

Assignment(s)

- *Due*: Final project proposal. | Canvas dropbox |
- *Due*: Exercise ZZ • Sleep & Circadian Rhythms. | Canvas dropbox |
- *Assigned*: Exercise EVO • Evolution. *Due*: 2026-03-19. | Canvas dropbox |

Spring Break: Mar 9-13

NO CLASS

Week 09: Mar 16-20

Thu, Mar 19

Topic(s)

- Development of the nervous system

Readings

- Required:
  - Cao, Huang, & He (2017)
  - Blumberg & Adolph (2023)
- Optional:
  - Larsen, Sydnor, Keller, Yeo, & Satterthwaite (2023)
  - Rakic (2009)

Materials

- Notes

Assignment(s)

- *Due*: Exercise EVO • Evolution. | Canvas dropbox |
- *Assigned*: Exercise DEV • Development. *Due*: 2026-03-26. | Canvas dropbox |

Week 10: Mar 23-27

Thu, Mar 26

Topics

- Cellular neuroscience I
  - Anatomy
  - Physiology
    - Resting potential

Readings

- Cellular neuroscience notes | PDF |
- Optional:
  - Won, Bhalla, Lee, & Lee (2025)
  - Zeng & Sanes (2017)
  - Oliveira, Sardinha, Guerra-Gomes, Araque, & Sousa (2015)
  - Distéfano-Gagné, Bitarafan, Lacroix, & Gosselin (2023)

Materials

- Slides | PDF |

Assignment(s)

- *Due*: Exercise DEV • Development | Canvas dropbox |
- *Assigned*: Exercise PHYS • Neurophysiology I. *Due*: 2026-04-02. | Canvas dropbox |

Week 11: Mar 30 - Apr 3

Thu, Apr 02

Topics

- Cellular neuroscience II
  - Action potential
  - Synaptic transmission

Readings

- Cellular neuroscience notes

Materials

- Slides

Assignment(s)

- *Due*: Exercise PHYS • Neurophysiology I | Canvas dropbox |
- *Assigned*: Exercise AP • Neurophysiology II. *Due*: 2026-04-09. | Canvas dropbox |

Week 12: Apr 6-10

Thu, Apr 09

Topics

- Neurochemistry
  - Neurotransmitters
  - Hormones
- Neurocomputing

Readings

- Neurochemistry notes
- Optional: Sarkar et al. (2016).

Materials

- Slides

Assignment(s)

- *Due*: Exercise AP • Neurophysiology II. | Canvas dropbox |
- *Assigned*: Exercise CHEM • Neurochemistry. *Due*: 2026-04-16. | Canvas dropbox |

Week 13: Apr 13-17

Thu, Apr 16

Topics

- Disorder & disease I

Readings

- Howes, Bukala, & Beck (2023)
- Volk, Chiu, Sharma, & Huganir (2015)

Materials

- Notes

Assignment(s)

- *Due*: Exercise CHEM • Neurochemistry. | Canvas dropbox |

Week 14: Apr 20-24

Thu, Apr 23

Topics

- Disorder & disease II

Readings

- Moncrieff et al. (2022)
- Namkung, Kim, & Sawa (2017)

Materials

- Notes

Week 15: Apr 27 - May 1

Thu, Apr 30

Topics

- Frontiers in the biology of behavior
- Beethoven and the Cerebral Symphony

Materials

- Slides

Finals: May 4-8

Thu, May 07

- Final Project write-up due. | Canvas dropbox |

References

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